Training a Playlist Curator Based on User Taste: Reproduce & Extend

ML Mini Project (UE24CS352A)

Team: 26

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1. Problem Statement

Given a song's Spotify audio features and genre tags, classify it into the playlist it belongs to (13-class classification). Playlist membership is subjective and not always separable, which makes this harder than plain genre classification. The project (a) reproduces the models and results reported in the reference paper and (b) extends it with one additional model not used there, evaluated on the same data, split and metrics.

2. Dataset Details

Playlists/tracks: the authors' exact 13 Spotify-curated playlists (1,027 track entries, Dec 2018 snapshot, from their public repository); one track listed in two playlists is removed, giving 1,025 tracks by 953 artists. Features: the authors' audio-feature and genre tables were never released and Spotify's audio-features API is closed to new apps, so we rebuilt them: the 10 audio features (danceability, energy, key, loudness, mode, speechiness, acousticness, liveness, valence, tempo) come from the Kaggle *Spotify Dataset 1921-2020* matched by track ID (248 tracks) or title + artist (132), and from the ReccoBeats API (298 tracks, estimates of Spotify's values). Genre tags are a multi-hot encoding of artist genres from the same Kaggle dataset. Tracks without features were dropped, leaving 678 tracks x 434 features (10 audio + 424 genre tags) versus the paper's 1,044 tracks and 2,095 tags. An 80/20 stratified split gives 542 training and 136 test songs.

3. Approach

Part 1, reproduction. We re-implemented every experiment in the paper: Perceptron and Polynomial/Sigmoid/RBF SVMs with and without standardisation (5-fold CV); an RBF SVM tuned on the penalty parameter C (and gamma) with per-playlist precision/recall/F1; and neural networks with one hidden layer (C sigmoid units) and two hidden layers (2C identity, then C sigmoid), trained with LogSoftmax output, NLL loss, L2 regularisation, full-batch gradient descent and 150 epochs. Hyper-parameters the paper does not state were tuned on a validation split of the training data only. Part 2, extension. We added XGBoost (gradient-boosted decision trees). The paper lists decision trees as future work, and trees suit this data: they need no feature scaling (which makes or breaks the SVMs and Perceptron here), handle dense audio plus sparse tag columns, and give feature importances. We also ran an ablation (audio-only vs. audio + tags) for all models.

5. Implementation Overview

Everything is in one Jupyter notebook (`playlist_curator.ipynb`) using Python, scikit-learn (SVM, Perceptron, pipelines, grid search, CV), PyTorch (neural networks) and XGBoost. The pipeline is: (1) download the authors' playlists and tracks; (2) attach audio features and genre tags and report coverage; (3) build the feature matrix; (4) create one fixed split shared by all models; (5) run Part 1; (6) run XGBoost with a grid over depth, learning rate and number of trees (3-fold CV inside the training set); (7) generate the *Reported vs. Yours* table, comparison plots, confusion matrices and ablation; (8) save all tables to `results/`. Scaling sits inside a Pipeline so it is fit only on training folds, and neural-network results are averaged over 5 seeds. A README gives setup and run instructions.

6. Results

Model	Reported (paper)	Ours: hold-out test	Ours: 5-fold CV
Perceptron (scaled)	0.76 (CV)	0.70	0.76 ± 0.02
Tuned RBF SVM	0.80 ± 0.05 (CV)	0.70	0.74 ± 0.04
NN, 1 hidden layer	0.82 test	0.75 ± 0.02 (5 seeds)	-
NN, 2 hidden layers	0.77 test	0.73 ± 0.01 (5 seeds)	-
XGBoost (new)	not in paper	0.735	0.77 ± 0.02

6 of 13 reported numbers are reproduced within 5 percentage points (the paper's own CV standard deviation), and all six qualitative claims hold on our data (scaling helps, tuning helps, the 1-hidden-layer network beats the SVM and the 2-layer network). Genre-specific playlists (Spread the Gospel, Celtic Punk, Country) are easy (F1 about 0.83-0.94); Kitchen Swagger is hard (F1 about 0.44-0.56).

7. Conclusions

- The paper's pipeline and its main conclusions are reproducible. Standardisation is essential for the Perceptron and SVMs, and a small one-hidden-layer network is the best reference model (0.75 vs. the paper's 0.82). Our numbers are 5-7 points lower for the tuned SVM and the network, mainly because the features were rebuilt and only 678 of 1,025 tracks could be matched.
- XGBoost performs similarly to the reference models (0.735 hold-out, 0.77 CV; differences are within about ±4 points of noise on 136 test songs). Its advantage shows on audio features alone (0.52 vs. 0.38 SVM and 0.44 NN). Genre tags carry about 97% of its importance, and with them every model reaches 0.70-0.76.
- Limitations: mixed exact and estimated audio features, a small test set, heavy overfitting (XGBoost train accuracy 1.00), and no access to the authors' private user playlists. Future work: node2vec/word2vec artist and tag embeddings, segmentation (at most 5 playlists at a time), and multi-label playlist membership.

Reference: Awadelkarim & Coelho, Stanford, 2018